
Supplementary information

Half-minute-scale atomic coherence and high relative stability in a tweezer clock

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Supplementary information: Half-minute-scale atomic coherence and high relative stability in a tweezer clock

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Appendix A: Alignment of optical potentials

1. Tweezer array alignment

The two tweezer arrays share a common reference - the microscope objective used to project them - and so alignment of the tweezers is both fairly straightforward and robust, with negligible drifts on the $0.5\ \mu\text{m}$ scale over the course of multiple days. To initially overlap the two tweezer arrays we align the imaging system to the 515 nm tweezers, and take destructive fluorescence images of atoms in the 813 nm tweezers. Varying the focus of the 813 nm system to bring the atoms into focus aligns the arrays in the axial direction. For fine alignment of the in-plane position of the 515 nm and 813 nm tweezer arrays, we scan the RF tones used to generate the 813 nm tweezers, optimizing for low atom loss and temperature after passing atoms between the two tweezer arrays.

2. Axial lattice alignment

The axial lattice is projected with an independent lens oriented perpendicular to the objective, and so does not share a convenient common reference with the tweezers. As such, both the position and angle of the fringes relative to the tweezer plane must be carefully optimized. The lattice beams are initially overlapped with the tweezers by loading atoms directly from the MOT into the lattice. Fluorescence images of these atoms are then used to center the lattice on the tweezer array. The axial alignment is coarsely optimized by measuring light shifts induced by the lattice on the $^1\text{S}_0 \leftrightarrow ^3\text{P}_1$ transition in non-magic magnetic fields.

To flatten the lattice fringes relative to the tweezers we parametrically heat atoms trapped in the axial lattice with the tweezers. Specifically, in the case that a tweezer is well centered on a lattice fringe, modulating the power in the lattice changes the overall trap depth experienced by an atom, but not the position of the trap center (Fig. S1b). This results in parametric heating with a resonance at $\omega_m = 2\omega_0$, where ω_0 is the trap frequency in the absence of

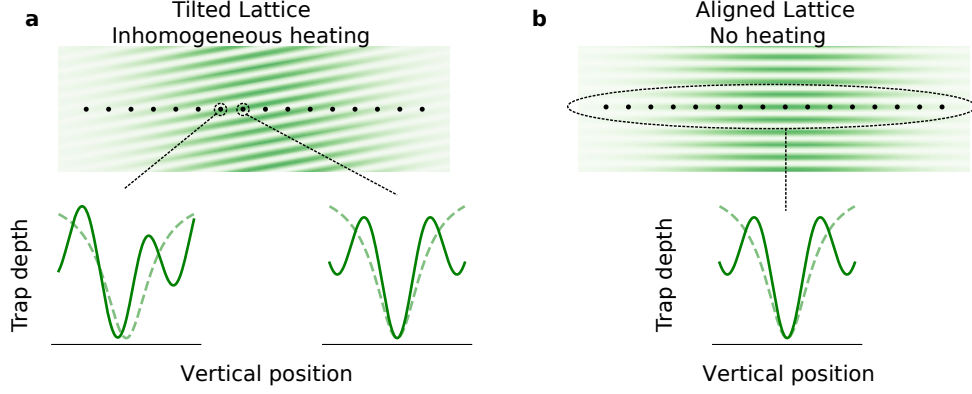


FIG. S1. Axial lattice alignment via parametric heating. a) The upper panel shows a side view of a misaligned axial lattice (green fringes) overlapped with the tweezer array (black dots). The lower plots show the tweezer potential (dashed curve) and the combined lattice and tweezer potential (solid green curve) at two locations. When the lattice fringe is tilted relative to the plane of the tweezers, it shifts the centers of the traps (as shown in the bottom left plot) in a position-dependent way. As such, modulating the depth of the tweezers will cause different parametric heating rates at different sites in the array. b) When the tweezers and lattice are well aligned, modulating the tweezers at the trap frequency ($\omega_m = \omega_0$) does not cause substantial heating.

modulation [1, 2]. In this case, if the modulation frequency is equal to the trap frequency, the atom is not strongly heated. If, on the other hand, the lattice and tweezer are misaligned, this modulation results in a shaking of the trap center, which can result in heating at $\omega_m = \omega_0$ (Fig. S1a) [1, 2]. For an appropriately chosen modulation amplitude and duration at $\omega_m = \omega_0$, we can use the observed probability of loss in a given tweezer as an indication of the relative alignment between that tweezer and the nearest lattice fringe. By scanning the phase of the lattice and fitting the phase of the resultant heating signal at each tweezer, we extract the relative orientation of the lattice and tweezers (as shown in extended data Fig. E1 of the main text). This signal allows for alignment of the lattice to the tweezers at the $\lambda_l/10$ level, where $\lambda_l \approx 1 \mu\text{m}$ is the lattice spacing.

Once optimized, this flattening has not been observed to change over multiple weeks. However, the spatial phase of the lattice can drift by a period over several hours due to a combination of thermal expansion and slightly mismatched path lengths in the lattice beams. This could eventually be addressed by actively feeding back on the lattice phase as measured by a camera, or by improving the passive stability of the lattice in a future design.

Appendix B: Model for light shifts in an optical tweezer potential

The geometry of an optical potential introduces important corrections to standard first order expressions for AC Stark shifts of atomic transitions. In this section, we follow the approach taken in [3], which studies light shifts in lattice potentials. We derive an analogous expression for optical tweezers (see also [4]), and isolate terms that are first-order-sensitive to the detuning of the trapping wavelength from the magic wavelength.

The intensity profile of an optical tweezer in cylindrical coordinates (z , $\rho = \sqrt{x^2 + y^2}$) is:

$$I(\rho, z) = I_0 [w_0/w(z)]^2 e^{-2\rho^2/w(z)^2}, \quad (\text{S2.1})$$

where I_0 is the peak intensity of the tweezer, and w_0 is the Gaussian beam waist at the focus. The waist as a function of the propagation distance z is $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$, where $z_R = \pi w_0^2/\lambda$ is the Rayleigh range. The optical potential is then $U(\rho, z) = \alpha^{E1}(\nu_T)I(\rho, z) + \beta(\nu_T)I(\rho, z)^2$, where $\alpha^{E1}(\nu_T)$ and $\beta(\nu_T)$ are the electric-dipole polarizability and hyperpolarizability, respectively, of ^{88}Sr trapped in a tweezer with optical frequency ν_T . The effects of magnetic-dipole and quadrupole polarizabilities in ^{88}Sr are not included in this analysis.

After expanding the potential $U(\rho, z)$ to fourth order in both ρ and z , we treat the quartic corrections as a perturbation to the harmonic solutions, and arrive at an approximate expression for the energy shift $\Delta E_{g,e}$ for the ground (g) and excited (e) electronic states. The resulting frequency shift ν_{LS} is expressed $\nu_{LS} = (\Delta E_e - \Delta E_g)/h$, where h is Planck's constant. For this work, we are interested in how ν_{LS} varies as a function of ν_T , as the inhomogeneities in ν_T across our tweezer array serve as a dominant source of dephasing between atoms.

In order to express our result in terms of the differential electric-dipole polarizability $\frac{1}{\alpha_m^{E1}} \frac{\partial \Delta \alpha^{E1}}{\partial \nu}$, where α_m^{E1} is the polarizability at the magic trapping frequency $\nu_m \simeq 369$ THz, and $\Delta \alpha^{E1}$ is the difference between the electric-dipole polarizabilities of the ground and excited states, we expand our result in powers of $\sqrt{I_0}$ up to I_0^2 . Including only terms that depend explicitly on the detuning of the tweezer frequency from the clock-magic trap frequency, $\delta_L = \nu_T - \nu_m$, the resulting equation for the differential light shift is

$$\frac{h\nu_{LS}}{E_R} \approx \left[\left(\sqrt{2} \frac{w_0}{z_R} (n_x + n_y + 1) + \left(\frac{w_0}{z_R} \right)^2 \left(n_z + \frac{1}{2} \right) \right) (\sqrt{u_e} - \sqrt{u_g}) - (u_e - u_g) \right] \quad (S2.2)$$

$$\approx \left[\frac{1}{2} \left(\sqrt{2} \frac{w_0}{z_R} (n_x + n_y + 1) + \left(\frac{w_0}{z_R} \right)^2 \left(n_z + \frac{1}{2} \right) \right) \sqrt{u_m} - u_m \right] \frac{1}{\alpha_m^{E1}} \frac{\partial \Delta \alpha^{E1}}{\partial \nu} \delta_L. \quad (S2.3)$$

In this expression, $u_{g,e} = \alpha_{g,e}^{E1}(\nu_T)I_0/E_R$ which, in going from Eqn. S2.2 to Eqn. S2.3, are Taylor expanded around $u_m = \alpha_m^{E1}I_0/E_R$. Finally, n_z , n_x , and n_y are the number of motional quanta in the axial direction, and two radial directions, respectively. Using the values in table E1 of the methods section, as well as $\frac{1}{\alpha_m^{E1}} \frac{\partial \Delta \alpha^{E1}}{\partial \nu} = -15.5(1.1) (\mu\text{Hz}/E_R)/\text{MHz}$ [5, 6], this calculation yields results that are within 3% of the values measured via ellipse fitting. Note that because there is substantially more than 3% error in the characterization of our trap depths and shape, we opt to use the measured values of these shifts to predict the expected Gaussian decay rate associated with dephasing which appears in the main text (see Fig. 2b of the main text).

Appendix C: Ellipse fitting

To extract the relative phase between two sub-ensembles in the array after a Ramsey measurement, we parametrically plot the excitation fraction in one sub-ensemble against the other. Fitting an ellipse to this distribution via its singular value decomposition, we extract values for the semi-major and minor axes of the best-fit ellipse, a , b . These are related to the relative phase between ensembles by $\phi = 2 \arctan(b/a)$.

We follow the procedure in [7] to calculate an expected uncertainty in ϕ due to QPN, with a slight modification to account for loss of Ramsey contrast due to atom loss as opposed to loss of coherence. Specifically, the excitation probability in the two ensembles are:

$$p_x = \frac{c_l}{2} (1 + c_d \cos \theta) \quad (S3.1)$$

$$p_y = \frac{c_l}{2} (1 + c_d \cos(\theta + \phi)),$$

where θ is the Ramsey phase of the first ensemble, and ϕ is the relative phase between the two ensembles — the quantity we are interested in measuring. In the regime where θ is well controlled, p_x and p_y can be thought of as the Ramsey signal associated with each sub-ensemble. We operate in the regime where atom-laser coherence is lost, such that θ is randomized. In this case it is still possible to extract ϕ from two ensembles that share a common θ , as we will see below. c_l is the contribution to the Ramsey contrast from loss (decay of the Ramsey signals towards zero), and c_d is the contribution to the Ramsey contrast from decoherence or dephasing (decay towards $c_l/2$). The overall Ramsey contrast is $c = c_l c_d$. To infer values for c_l and c_d from the measured rate of decay of c , we fit the c as a function of time to the product of an exponential and a Gaussian. The Gaussian component is presumed to be exclusively due to dephasing, and thus contributes only to c_d . The exponential component is broken up between the theoretically known ratios between processes that remove atoms from the $\{|e\rangle, |g\rangle\}$ manifold, namely atom loss and Raman scattering from 3P_0 to 3P_2 , and processes that project onto $|e\rangle$ or $|g\rangle$, namely black body or Raman scattering from 3P_0 to 3P_1 , and black body scattering from 3P_0 back to 3P_0 . The former processes contribute to c_l , and the latter to c_d .

We now define the quantities x and y , which correspond to the measured excitation fraction in the two ensembles offset to be centered at 0:

$$x = \frac{c}{2} \cos \theta \quad (S3.2)$$

$$y = \frac{c}{2} \cos(\theta + \phi).$$

Note that while these quantities depend only on the product of c_l and c_d , this is not true of their variance, which can be written as:

$$\begin{aligned}\sigma^2(x) &= \frac{1}{N}p_x(1-p_x) \\ &= -\frac{c_l}{4N}(1+c_d\cos(\theta))(c_l-2+c_dc_l\cos(\theta)),\end{aligned}\tag{S3.3}$$

where $N \sim 75$ is the number of atoms in each ensemble. An analogous expression holds for $\sigma^2(y)$. Based on this, we can infer a variance in our measurement of ϕ :

$$\begin{aligned}\sigma^2(\phi) &= \left|\frac{\partial\phi}{\partial x}\right|^2\sigma^2(x) + \left|\frac{\partial\phi}{\partial y}\right|^2\sigma^2(y) \\ &= \frac{4}{c^2}(\csc^2(\theta)\sigma^2(x) + \csc^2(\theta+\phi)\sigma^2(y)).\end{aligned}\tag{S3.4}$$

Because we operate with Ramsey dark times far longer than the atom-laser coherence time, we effectively average over all Ramsey phases θ . In this case:

$$\langle\sigma^2(\phi)\rangle = \frac{4}{c^2}\left(\int_0^{2\pi}\frac{d\theta}{2\pi}\frac{1}{\csc^2(\theta)\sigma^2(x) + \csc^2(\theta+\phi)\sigma^2(y)}\right)^{-1},\tag{S3.5}$$

In the limiting case where $c = 1$, QPN results in a variance of $\langle\sigma^2(\phi)\rangle = 2/(Nc^2)$ as expected. For $c < 1$ and as ϕ approaches $\pi/2$ however, $\langle\sigma^2(\phi)\rangle$ increases by an additional scale factor.

This would suggest that to maximize sensitivity it would be beneficial to operate near $\phi = 0$, however, as stated in the main text, the ellipse fitting procedure is biased near this phase. This is due to a combination of two effects: first, these fits cannot distinguish between positive and negative values of ϕ , and only return $\phi \geq 0$. Second, the effective probability distribution the data is sampled from is the convolution of an ellipse and a Gaussian distribution associated with QPN. As a result, for $\phi \simeq 0$, data points can cross the semi-major axis due to QPN. In this case, due to the first effect, instead of cancelling with points that cross over from the other side, this results in a weighting away from $\phi = 0$. Based on this reasoning, a practical requirement is that ϕ is sufficiently large for two peaks to be resolvable along a slice through the semi-minor axis of the ellipse. A similar line of reasoning holds for very long interrogation times, where the contrast of the signal becomes small enough that points can cross the semi-minor axis of the ellipse due to QPN.

To quantify the effect of this biasing, we generate artificial data with a known contrast and phase offset via Monte-Carlo simulations that capture the QPN and measured decoherence in our system. By comparing the phase and contrast extracted via ellipse fitting to the known parameters used to generate the data, it is possible to extract a fractional error due to biasing (Fig. S2). For measuring the decay of Ramsey contrast, we use this error model to correct the data and provide a bias-free estimate of the contrast. For measurements of the phase, we are simply interested in providing bounds on this error to ensure that we operate in an unbiased regime. At $\phi > 0.65$ we can bound this error to below 10%, which is the regime in which we perform the measurement of relative stability in the main text.

To calculate a variance in the extracted value of ϕ we turn to the jackknifing procedure described in [7]. Specifically, in order to compute an Allan variance it is necessary to have a single-shot estimate of the phase. This is done by checking how much the single-shot measurement of interest, ϕ_i , sways the estimated phase of the whole distribution:

$$\phi_i = n\phi - (n-1)\phi_{\neq i},\tag{S3.6}$$

where n denotes the total number of samples, and $\phi_{\neq i}$ is the estimated phase of the distribution with the i^{th} sample removed. By performing this analysis across all i the Allan variance can be computed in the normal way. The variance in ϕ can also be approximated as:

$$\sigma^2(\phi) = \frac{n-1}{n}\sum_i(\phi - \phi_{\neq i})^2.\tag{S3.7}$$

This analysis is used on real data to determine an optimal dark time in the main text, but also on simulated data to calculate a correction factor to the statistical variance of the biased fits compared to QPN. For the conditions of the relative stability measurement in the main text, at $\phi = 0$ this results in a correction factor of 0.70(1). For $\phi > 0.65$ this analysis yields results bounded to within 5% of the correct value. This extra source of error is included in the error bars on the relative fractional frequency stability quoted in the main text. We also confirm that including the fringes observed in Fig. 4a of the main text does not alter the results of this analysis — simulated data with and without these fringes yield results that are within error bars of each other, as is clarified in supplement section E.

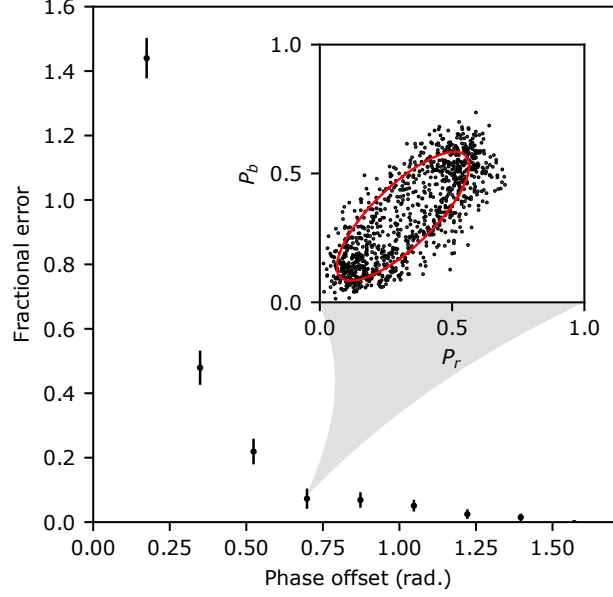


FIG. S2. Quantifying bias in ellipse fitting. By simulating data with a known phase offset, we can compute the fractional error in the phase extracted via ellipse fitting. Callout shows an example of a simulated data set with conditions similar to those used for the measurement of relative stability in the main text, where this error is bounded to below 10%.

Appendix D: Measuring atomic coherence with the correlation function

In this section, we define the correlator used in the main text, and show how it can be used to quantify atomic coherence. We will use the ordered basis $\{|e\rangle, |g\rangle, |0\rangle\}$ to describe the state of each atom, where $|e\rangle$ and $|g\rangle$ correspond to the 3P_0 and 1S_0 states respectively, and $|0\rangle$ corresponds to either a dark state (e.g. 3P_2), or the absence of an atom in a given tweezer. Our measurements do not distinguish between $|g\rangle$ and $|0\rangle$, such that we are always measuring the observable:

$$\sigma^z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (\text{S4.1})$$

Given these definitions, we define the correlation function between the measured state of two atoms in tweezers i and j as:

$$\begin{aligned} g_{ij}^{(2)} &= \langle (\sigma_j^z - \langle \sigma_j^z \rangle) \otimes (\sigma_i^z - \langle \sigma_i^z \rangle) \rangle \\ &= \langle \sigma_j^z \otimes \sigma_i^z \rangle - \langle \sigma_j^z \rangle \langle \sigma_i^z \rangle, \end{aligned} \quad (\text{S4.2})$$

where the angled brackets denote expectation values. In practice, these brackets denote an average over runs of the experiment that, after initial loading, included an atom in both sites i and j . This ensures that fluctuations in the number or distribution of tweezers that are filled with atoms only affect the number of samples and not the value

of the correlator. Note that given our fill fraction of ~ 0.45 , the probability of any given pair being populated on a single shot is $(0.45)^2 \simeq 0.2$.

In the main text, we are interested in averaging $g_{ij}^{(2)}$ over different ensembles $\{i, j\}$ of interest. Specifically, we define $C(\Delta\vec{r})$ as the average of $g_{ij}^{(2)}$ over all atom pairs $\{i, j\}$ that are separated by a displacement vector $\Delta\vec{r}$. Similarly, we define $C_{n \times n}$ as the average of $g_{ij}^{(2)}$ over all pairs of atoms in an $n \times n$ block, also averaged over all such blocks contained in the array for improved statistics. Note that in the limit of an infinitely large array (i.e. ignoring edge effects), $C_{n \times n} = \langle C(\Delta\vec{r}) \rangle_{|\Delta r_x|, |\Delta r_y| \leq n}$. In the following sections, we will make explicit the connection between the measured quantities defined above, and the off-diagonal coherences in the single atom density matrix.

Ramsey pulse and coherent evolution

To begin, we consider the evolution of atoms under the Ramsey sequence employed in our experiment. After initial loading, a tweezer j containing an atom has the density matrix $\rho_j^0 = |g\rangle\langle g|$. We perform the first of two SU(2) rotations by applying our clock laser to the atoms. This can be represented by the unitary operator U_1 :

$$U_1 = \begin{pmatrix} \cos(\alpha/2) & -i \sin(\alpha/2) & 0 \\ -i \sin(\alpha/2) & \cos(\alpha/2) & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (\text{S4.3})$$

where we have chosen a reference frame in which the rotation occurs about the x -axis; and α is the rotation angle, which would be $\pi/2$ in an ideal sequence. This transforms the system into the state $\rho_j^1 = U_1 \rho_j^0 U_1^\dagger$. The atom is then held in this state for a variable dark time. Due to the energy splitting between $|g\rangle$ and $|e\rangle$, this applies the unitary operator U_j^{hold} :

$$U_j^{\text{hold}} = \begin{pmatrix} e^{i\phi_j/2} & 0 & 0 \\ 0 & e^{-i\phi_j/2} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (\text{S4.4})$$

This evolution is indexed by the tweezer site j in order to account for spatially inhomogeneous light shifts. Assuming perfect unitary evolution, this leaves that atom in the state ρ_j^{R} :

$$\rho_j^{\text{R}} = \begin{pmatrix} \sin^2(\alpha/2) & -ie^{i\phi_j} \cos(\alpha/2) \sin(\alpha/2) & 0 \\ ie^{-i\phi_j} \cos(\alpha/2) \sin(\alpha/2) & \cos^2(\alpha/2) & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (\text{S4.5})$$

However, throughout this sequence the atom can also be projected into other states via the scattering and loss mechanisms discussed in other sections. Since these processes are either incoherent, or drive the atom into the $|0\rangle$ state, we can model their effects by taking the state at the end of this sequence to be a statistical mixture of ρ_j^{R} and the three basis states. This new state is defined as $\tilde{\rho}_j$:

$$\tilde{\rho}_j = p^{(\text{R})} \rho_j^{\text{R}} + p^{(g)} |g\rangle\langle g| + p^{(e)} |e\rangle\langle e| + p^{(0)} |0\rangle\langle 0|, \quad (\text{S4.6})$$

where $p^{(\text{R})}$, $p^{(e)}$, $p^{(g)}$, and $p^{(0)}$ are the classical probabilities associated with each of these states, and sum to 1. We label the off diagonal of the single-particle density matrix, $|\rho_{eg}|$, as the single-atom coherence. For perfect $\pi/2$ pulses α , $2|\rho_{eg}| = p^{(\text{R})}$ is equal to the Ramsey contrast c . In the limit of no dephasing, the decay of the ensemble average of $|\rho_{eg}|$ or $p^{(\text{R})}$ captures the rate at which decoherence occurs due to projection onto any of the states $|0\rangle$, $|g\rangle$, or $|e\rangle$, from atom loss, spontaneous emission, BBR, or Raman scattering.

Measurement protocol

In order to complete the Ramsey sequence, we apply another SU(2) rotation. However, for long hold times the loss of atom-laser coherence means that this rotation occurs about a random axis on the equator of the Bloch sphere. This rotation, which we call U_2 , can be written as:

$$U_2 = \begin{pmatrix} \cos(\alpha/2) & -ie^{-i\theta} \sin(\alpha/2) & 0 \\ -ie^{i\theta} \sin(\alpha/2) & \cos(\alpha/2) & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (\text{S4.7})$$

where θ parametrizes the rotation axis $\hat{n} = \cos(\theta)\hat{x} + \sin(\theta)\hat{y}$. This equation assumes that the second rotation sweeps out the same angle α as the first. The final density matrix that we measure, $\rho_j = U_2 \tilde{\rho}_j U_2^\dagger$, has five non-zero elements:

$$\rho_j^{00} = p^{(R)} [2 \sin^2(\alpha/2) \cos^2(\alpha/2) (\cos(\phi_j + \theta) + 1)] + p^{(e)} \cos^2(\alpha/2) + p^{(g)} \sin^2(\alpha/2) \quad (\text{S4.8})$$

$$\begin{aligned} \rho_j^{01} = & ie^{-i\theta} [p^{(R)} \sin^3(\alpha/2) \cos(\alpha/2) (1 + e^{-i(\phi_j + \theta)}) - p^{(R)} \cos^3(\alpha/2) \sin(\alpha/2) (1 + e^{i(\phi_j + \theta)}) \\ & + \sin(\alpha/2) \cos(\alpha/2) (p^{(e)} - p^{(g)})] \end{aligned}$$

$$\begin{aligned} \rho_j^{10} = & -ie^{i\theta} [p^{(R)} \sin^3(\alpha/2) \cos(\alpha/2) (1 + e^{i(\phi_j + \theta)}) - p^{(R)} \cos^3(\alpha/2) \sin(\alpha/2) (1 + e^{-i(\phi_j + \theta)}) \\ & + \sin(\alpha/2) \cos(\alpha/2) (p^{(e)} - p^{(g)})] \end{aligned}$$

$$\begin{aligned} \rho_j^{11} = & p^{(R)} [\sin^4(\alpha/2) + \cos^4(\alpha/2) - 2 \cos^2(\alpha/2) \sin^2(\alpha/2) \cos(\phi_j + \theta)] \\ & + p^{(g)} \cos^2(\alpha/2) + p^{(e)} \sin^2(\alpha/2) \end{aligned}$$

$$\rho_j^{22} = p^{(0)}.$$

The main quantity we are interested in measuring is the correlator $g_{ij}^{(2)}$, for which we need the density matrix associated with the full tweezer-array. Since the atoms are not entangled, this is simply $\rho = \bigotimes_{i=0}^{N-1} \rho_i$, where N is the number of tweezers in the array.

To write down an expression for $g_{ij}^{(2)}$ we first compute $\langle \sigma_j^z \rangle_\theta$, where θ serves as a reminder that we are really sampling from a compound probability distribution in which θ is a uniformly distributed random variable. In this case we have:

$$\begin{aligned} \langle \sigma_j^z \rangle_\theta &= \rho_j^{00} - \rho_j^{11} - \rho_j^{22} \\ &= p^{(R)} [\sin^2(\alpha) \cos(\phi_j + \theta) - \cos^2(\alpha)] + (p^{(e)} - p^{(g)}) \cos(\alpha) - p^{(0)}. \end{aligned} \quad (\text{S4.9})$$

Marginalizing over θ we have:

$$\begin{aligned} \langle \sigma_j^z \rangle &= \frac{1}{2\pi} \int_0^{2\pi} \langle \sigma_j^z \rangle_\theta d\theta \\ &= (p^{(e)} - p^{(g)}) \cos(\alpha) - p^{(R)} \cos^2(\alpha) - p^{(0)}. \end{aligned} \quad (\text{S4.10})$$

Because ρ is a tensor product of the single particle density matrices, we also have $\langle \sigma_j^z \otimes \sigma_i^z \rangle_\theta = \langle \sigma_j^z \rangle_\theta \langle \sigma_i^z \rangle_\theta$, giving:

$$\langle \sigma_j^z \otimes \sigma_i^z \rangle = [p^{(R)}]^2 \sin^4(\alpha) \frac{\cos(\phi_j - \phi_i)}{2} + \langle \sigma_j^z \rangle \langle \sigma_i^z \rangle. \quad (\text{S4.11})$$

Plugging Eqns. S4.10 and S4.11 into Eqn. S4.2 yields:

$$g_{ij}^{(2)} = \frac{1}{2} [p^{(R)}]^2 \sin^4(\alpha) \cos(\phi_j - \phi_i). \quad (\text{S4.12})$$

Expectation value of measurement protocol

Restricting our attention to the measurement of $C_{n \times n}$ as defined above, we consider the average correlator between all non-empty tweezers within a block b :

$$C_b \equiv \frac{1}{N(N-1)} \sum_{i \neq j} g_{ij}^{(2)}, \quad (\text{S4.13})$$

where N is the number of atoms in the block. The average over trials should converge to the value $g_{ij}^{(2)}$ calculated in the previous section, giving:

$$C_b = \frac{[p^{(R)}]^2 \sin^4(\alpha)}{2N(N-1)} \sum_{i \neq j} \cos(\phi_j - \phi_i). \quad (\text{S4.14})$$

Note that $C_{n \times n}$ is simply the average of C_b over all blocks in the array of the appropriate class, and, to the extent that each block behaves identically, will simply equal C_b .

In addition to looking at the correlator between all pairs within a block, we also look at the correlator between atoms loaded into a single tweezer site, and all of the other atoms in the array (Fig. 4b in main text). If the single atom has index j , then the outcome of this analysis yields

$$C_{j,A} = \frac{[p^{(R)}]^2 \sin^4(\alpha)}{2(N-1)} \sum_{i \in A} \cos(\phi_i - \phi_j), \quad (\text{S4.15})$$

where A contains all indices $i \neq j$ in a reference ensemble of interest. We next define the average phase for the atoms in the ensemble A as ϕ_A , and $\Delta\phi_i \equiv \phi_i - \phi_A$. Given that the dominant contribution to dephasing is from tweezer-induced frequency shifts, which are symmetrically distributed about the anti-diagonal of the array, we can assume that the $\Delta\phi_i$'s are symmetrically distributed about zero. In this case:

$$C_{j,A} = \frac{[p^{(R)}]^2 \sin^4(\alpha)}{2(N-1)} \cos(\phi_A - \phi_j) \sum_{i \in A} \cos(\Delta\phi_i). \quad (\text{S4.16})$$

For large atom number where A is indistinguishable from the full array, $p^{(R)} \sin^2(\alpha) \sum_{i \in A} \cos(\Delta\phi_i)/N$ is equal to the Ramsey contrast c , including the effects of imperfect $\pi/2$ pulses and dephasing. c can be independently characterized using the ellipse fitting methods discussed in supplemental section C, and divided out of Eqn. S4.16 (as is done in Fig. 4b of the main text). For atoms near the center of the array where $\phi_j \simeq \phi_A$, this yields a direct measurement of $p^{(R)} \sin^2(\alpha)$, which is insensitive to dephasing and proportional to the single-atom coherence $|\rho_{eg}|$. Note that if $\phi_j \neq \phi_A$, this measurement decays strictly faster than $|\rho_{eg}|$. In practice, we average $C_{j,A}$ over a 4×4 region at the center of the array, yielding C_A . To extract the single-atom coherence from C_A , we fit it with a fit function that includes a fixed contribution from the decay of c extracted from ellipse fitting, and a free exponential component. The $1/e$ lifetime of this free component is the inferred single-atom coherence time, and has a value of 48(8) s.

Comparison with the average density matrix

It is possible to show that a measurement of $\sqrt{C_{n \times n}}$ constitutes a lower bound on the atomic coherence. To begin, consider the average density matrix $\bar{\rho}$ of all atoms in a given block:

$$\bar{\rho} \equiv \frac{1}{N} \sum_{j=0}^{N-1} \tilde{\rho}_j = \begin{pmatrix} p^{(e)} & 0 & 0 \\ 0 & p^{(g)} & 0 \\ 0 & 0 & p^{(0)} \end{pmatrix} + \frac{p^{(R)} \sin(\alpha)}{2} \begin{pmatrix} \tan^2(\alpha/2) & \frac{1}{N} \sum_{j=0}^{N-1} -ie^{i\phi_j} & 0 \\ \frac{1}{N} \sum_{j=0}^{N-1} ie^{-i\phi_j} & \cot^2(\alpha/2) & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (\text{S4.17})$$

We define the atomic coherence as the magnitude of the off-diagonal element, $|\bar{\rho}_{eg}|$, where:

$$\begin{aligned}
|\bar{\rho}_{eg}|^2 &= \frac{(p^{(R)} \sin(\alpha))^2}{4} \frac{1}{N} \sum_{j=0}^{N-1} e^{-i\phi_j} \frac{1}{N} \sum_{j'=0}^{N-1} e^{i\phi_{j'}} \\
&= \left(\frac{p^{(R)} \sin(\alpha)}{2N} \right)^2 \left[\sum_{j,j'} \cos(\phi_j - \phi_{j'}) \right].
\end{aligned} \tag{S4.18}$$

Writing this in terms of C_b using Eqn. S4.14 yields:

$$|\bar{\rho}_{eg}|^2 = \frac{1}{N} \left(\frac{p^{(R)} \sin(\alpha)}{2} \right)^2 + \frac{1}{\sin^2(\alpha)} \frac{C_b}{2}, \tag{S4.19}$$

which then gives:

$$C_b \leq 2 \sin^2(\alpha) |\bar{\rho}_{eg}|^2 \leq 2 |\bar{\rho}_{eg}|^2. \tag{S4.20}$$

The factor of 2 in this inequality is due to the differing normalizations of $|\bar{\rho}_{eg}|$ and $C_{n \times n}$. Specifically, $|\bar{\rho}_{eg}|^2$ ranges from 0 to 1/4, whereas after marginalizing over the laser phase, $C_{n \times n}$ varies between $-1/2$ and $1/2$ (where we are only concerned with the limit in which phase shifts are small, and $C_{n \times n} > 0$).

Eqn. S4.20 implies that a measurement of $\sqrt{C_b/2}$ is a lower bound on the atomic coherence, since $\sqrt{C_b/2} \leq |\bar{\rho}_{eg}|$. It is natural to average $\sqrt{C_b/2}$ over all appropriate b in the array, using $\langle \sqrt{C_b/2} \rangle$ to characterize the array-averaged atomic coherence. However, to avoid negative values in the square root due to statistical fluctuations we instead average over b before performing the square root, yielding $\sqrt{C_{n \times n}/2}$ where $C_{n \times n} = \langle C_b \rangle$. Note that by the triangle inequality $\sqrt{C_{n \times n}/2} \leq \langle \sqrt{C_b/2} \rangle$, and still constitutes a valid lower bound. Assuming that α does not vary with dark time, the lifetime of $\sqrt{C_{n \times n}}$ similarly provides a lower bound on the atomic coherence time.

Appendix E: Fringes and spatially varying frequency shifts

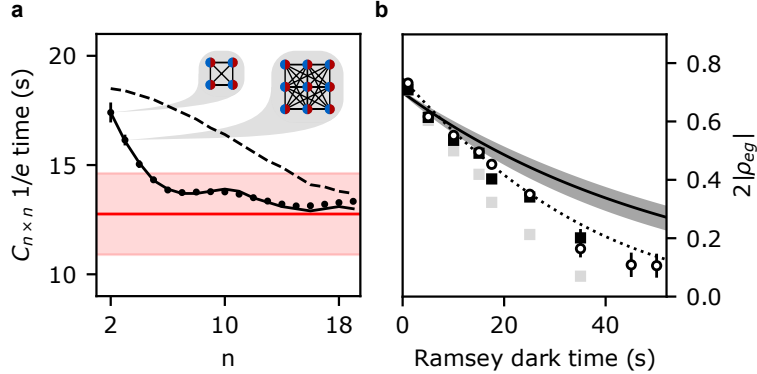


FIG. S3. Decay of correlations as a function of sub-ensemble size. a) Extending the analysis of correlations in 2×2 blocks of atoms in the main text to $n \times n$ blocks of varying size (see callouts) reveals that the inferred atomic coherence decays faster with increasing block size than the prediction based exclusively on theoretical tweezer frequency shifts and a fitted exponential decay constant (dashed line). An extended model that includes the fringes that appear in the correlator (solid line) is in good agreement with the data. The inferred lifetime based on the ellipse fitting procedure in Fig. 3a of the main text is shown in red (shaded region denotes 1σ confidence interval) and is in good agreement with this analysis. Note that for $n > 16$, the blocks are rectangular with dimensions $n \times 16$. b) The bound on atomic coherence based on a measurement of $C_{n \times n}$ becomes looser with larger n due to increased sensitivity to dephasing. For example, the atomic coherence inferred from averaging over all correlators, or equivalently $C_{20 \times 16}$ (grey squares), decays much more quickly than for $C_{2 \times 2}$ (open circles, reproduced from Fig. 4c of the main text). However, based on our knowledge of this dephasing we can remove the contribution from dephasing from the atomic coherence inferred from $C_{20 \times 16}$, which yields results (black squares) that are in good agreement with the bound set by $C_{2 \times 2}$.

We observe fringes in $C(\Delta\vec{r})$ that grow linearly in amplitude as a function of dark time. To study the effect of these fringes on our measurements we compute the rate at which correlations decay in $n \times n$ sub-ensembles of varying size (see Fig. S3a). For intermediate block sizes, the presence of the fringes results in substantially more rapid decay of the correlator. However, for block sizes that are small compared to the fringes, as is the case for the 2×2 blocks used in the main text, and for large blocks that sample the entire array, the presence of these fringes only marginally increases the decay rate. As a result these fringes do not have a substantial impact on either our ability to extract a single-atom coherence, or on the stability achievable in a synchronous clock comparison employing the entire array. Critically, the excess decay when averaging over the whole array compared to the single-atom coherence is fully explained by dephasing. This is further evidenced by the fact that the 20×16 correlator is in good agreement with the 2×2 correlator when it is corrected for the expected reduction in correlations due to dephasing at each time (Fig. S3b).

The relatively fine spatial structure of these fringes suggests that they are an optical effect, and the fact that they grow with dark time suggest that they are due to light shifts caused by the 813 nm tweezers, since all other beams are off during this period. This is further evidenced by the fact that the fringes are oriented along the more tightly spaced axis of the array, which leads to greater overlap between the tweezers and difficulties reading out, and thus balancing, their depths. Given the known $\sim 10\%$ inhomogeneities in trap depth in our system, a 200 MHz detuning of the tweezers from the magic condition can explain this effect. Moreover, the magnitude of this effect fluctuates day to day, which is consistent with the fact that the trap laser frequency is known to drift on the 50 MHz scale.

Appendix F: Master equation modeling of atom-atom correlations

In this section we introduce a full master equation-based model of our system, which we compare to our measurements of $C_{2 \times 2}$. Here, besides the $|e\rangle$, $|g\rangle$, and $|0\rangle$ states introduced in section D, we also include four motional states of the atom to account for finite atomic temperature. As a result, the state of each atom is represented by a 12×12 density matrix ρ . To compare to our measurements of $g^{(2)}$, we simulate two such atoms. In this case, because there is no mechanism for the development of quantum correlations, each atom can be simulated independently.

The atoms begin in a statistical mixture of $|g\rangle$ and $|0\rangle$ due to a total of 9% loss during the imaging and hand-off from the auxiliary to the science potential. To simulate our Ramsey measurement, each atom is rotated in the $\{|e\rangle, |g\rangle\}$ subspace by an angle α , which is $\pi/2$ for the average thermal ensemble. Specifically, each atom samples a thermal Boltzmann distribution defined by an average motional excitation number of $\bar{n} = 0.14$, and the rotation angle is modified according to each atom's sampled motional excitation state n (see supplement G). The atoms then undergo free evolution, accruing a phase θ relative to the laser. They further experience incoherent decay, and coherent dephasing which yields a relative phase ϕ . For readout, the result is rotated as at the beginning of the sequence, although n and thus α can be modified by heating due to Rayleigh scattering. With this final state in hand, we can compute expectation values of the relevant operators to obtain our results.

The evolution of the density matrix ρ is given by a Lindblad master equation

$$\dot{\rho} = -i[H, \rho] + \sum_i \Gamma_i \mathcal{L}(\rho, \mathcal{P}_i) \quad (\text{S6.1})$$

$$\mathcal{L}(\rho, \mathcal{P}) = \mathcal{P}\rho\mathcal{P}^\dagger - \frac{1}{2}(\rho\mathcal{P}^\dagger\mathcal{P} + \mathcal{P}^\dagger\mathcal{P}\rho), \quad (\text{S6.2})$$

where H is the Hamiltonian, Γ_i is the rate of the i^{th} incoherent process, and $\mathcal{P}_i$ is the corresponding jump operator. To write down expressions for H and $\mathcal{P}_i$, we define the single-atom Pauli matrices, which operate on the ground-excited subspace of the internal degrees of freedom:

$$\sigma_x^0 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \otimes \mathbb{1}_4, \quad \sigma_y^0 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \otimes \mathbb{1}_4, \quad \sigma_z^0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \otimes \mathbb{1}_4 \quad (\text{S6.3})$$

where $\mathbb{1}_4$ is the 4×4 identity matrix on the motional degrees of freedom.

To compare to our measurements of $C_{2 \times 2}$, we simulate pairs of atoms with a differential clock frequency shift that is equal to the mean tweezer-induced frequency difference between atom pairs in a 2×2 block. In this case the two atoms evolve under the Hamiltonians $H = \pm\delta_1\sigma_z^0/2$, where $\delta_1 = 2\pi \times 2.4$ mHz (see supplement B). Note that δ_1 includes the effects of the fringes observed in the correlator (see Fig. 4a in the main text, and Fig. S3).

The values of Γ_i are inferred from the measured decay rates presented in extended data table E2, and the theoretically known relative rates and branching ratios between the various relevant processes. Specifically, given the

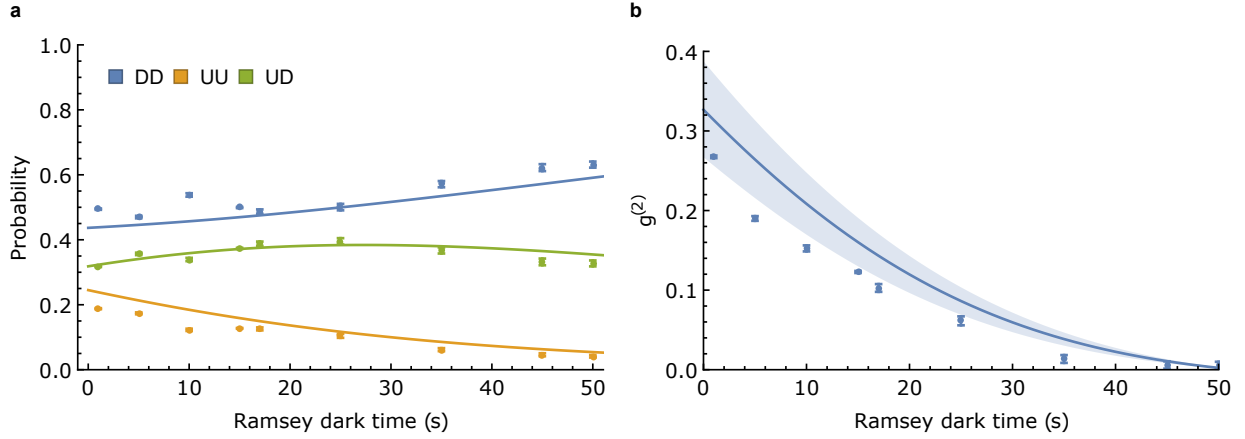


FIG. S4. Results of the master equation model. (a) The observed population ratios of UU, UD, and DD are in reasonable agreement with the prediction of our model with no free parameters. (b) We compare the observed 2×2 block correlator, $C_{2 \times 2}$ (blue points, see also Fig. 4 of the main text) to the prediction of our model with no free parameters. This model predicts a correlation $1/e$ time of 19.7(1.2) seconds, somewhat longer than the observed $1/e$ time of 17.4(9) seconds (this value includes a Gaussian and an exponential component, and is quoted in the main text as 33(2) s, associated with the decay of $\sqrt{C_{2 \times 2}}$). The errorband propagates uncertainty in the 3P_0 state lifetime and initial temperature; the uncertainty in the ground state lifetime is negligible. If we do not assume that all three axes thermalize after Rayleigh heating, such that $\Gamma_H = \Gamma_R^0/3$, the model's $1/e$ time increases by 5% while the initial value remains approximately constant.

measured excited-state decay rate Γ_e , loss rate of excited-state atoms Γ_e^t , and independently characterized black body decay rate Γ_1^{BBR} [8], the total Raman scattering rate due to the science potential is $\Gamma_{12}^R = \Gamma_e - \Gamma_e^t - \Gamma_1^{\text{BBR}}$. Given the relative rates of the Raman (and Rayleigh) scattering processes of $R_J^S = 0.416, 0.372$, and 0.212 for $J = 0, 1$, and 2 (where these rates are normalized to the total scattering rate), we can infer the individual process rates $\Gamma_J^R = \Gamma_{12}^R R_J^S / (R_1^S + R_2^S)$.

Based on the above definitions, we incorporate decay from the excited to ground state by setting $\Gamma = \Gamma_1^R + \Gamma_1^{\text{BBR}} = 8.7 \cdot 10^{-3} \text{ s}^{-1}$ and the corresponding jump operator to be $\sigma^- = \frac{1}{2}(\sigma_x^0 - i\sigma_y^0)$. We incorporate atomic loss from the ground and excited states by setting $\Gamma_g^L = \Gamma_g^t = 9.9 \cdot 10^{-3} \text{ s}^{-1}$ and $\Gamma_e^L = \Gamma_e^t + \Gamma_2^R = 14.6 \cdot 10^{-3} \text{ s}^{-1}$, with the corresponding jump operators $\sigma_{g0} = |0\rangle\langle g| \otimes \mathbb{1}_4$ and $\sigma_{e0} = |0\rangle\langle e| \otimes \mathbb{1}_4$. We incorporate blackbody Rayleigh scattering off the excited state driving the $^3P_0 \rightarrow ^3D_1 \rightarrow ^3P_0$ process by using $\Gamma_0^{\text{BBR}} = \Gamma_1^{\text{BBR}} R_0^D / R_1^D = 3.45 \cdot 10^{-3} \text{ s}^{-1}$ and the corresponding jump operator to be $\sigma_{ee} = |e\rangle\langle e| \otimes \mathbb{1}_4$. Finally, we incorporate heating due to Rayleigh scattering of the 813 nm tweezer light by setting $\Gamma^H = 2\Gamma_0^R/3 = 4.8 \cdot 10^{-3} \text{ s}^{-1}$ and the corresponding jump operator to be $\sigma_H = \mathbb{1}_3 \otimes \exp(i\eta(a + a^\dagger))$, where a ($a^\dagger$) is the motional lowering (raising) operator. The factor of $2/3$ accounts for two photons scattering per event and all axes thermalizing. Note that Rayleigh scattering of the tweezer light does not directly introduce dephasing since the scattered photon carries no information about which internal state the atom was in [8].

We numerically solve this master equation for each atom, and run the simulation for ten different relative phase shifts θ uniformly distributed between 0 and 2π to account for experimental averaging over the laser phase; we find that increasing the number of laser phases simulated does not affect our results.

As a function of time, we compute the average $g^{(2)}$ correlator as well as the expectation values of other observables corresponding to experimentally accessible two atom populations. Each run of our experiment returns a pair of images, which may be binarized into a pair of 20×16 arrays of ones and zeros corresponding to the presence or lack of an atom respectively. The first image corresponds to all atoms present in the array before the first $\pi/2$ pulse, while the second image shows atoms in the excited clock state after the second $\pi/2$ pulse. We do not distinguish between an atom in the ground state and the atom being lost. Therefore, we construct the observables

$$UU = P_e \otimes P_e \quad (\text{S6.4a})$$

$$UD = P_e \otimes (P_g + P_0) + (P_g + P_0) \otimes P_e \quad (\text{S6.4b})$$

$$DD = (P_g + P_0) \otimes (P_g + P_0), \quad (\text{S6.4c})$$

where the projectors are $P_e = |e\rangle\langle e|$, $P_g = |g\rangle\langle g|$, and $P_0 = |0\rangle\langle 0|$. As a test of our master equation model, in Fig. S4a we compare the expectation values of these three observables between the model and the experimental data, finding close agreement with no free parameters. However, the model slightly overestimates the decay time and initial

value of the correlator $g^{(2)}$ (Fig. S4b). This reflects that the correlator is much more sensitive to model parameters than any of the pair population results because it is equal to a difference between quantities of similar magnitude, $g^{(2)} = \langle UU \rangle + \langle DD \rangle - \langle UD \rangle - \langle \sigma_z \rangle_i \langle \sigma_z \rangle_j$ (where i and j label the two atoms under measurement, and σ_z is as defined in section D). Nevertheless, this discrepancy could indicate slightly higher decay rates than independently calibrated, higher atomic temperature of around $\bar{n} = 0.25$, increased imaging or handoff loss, or some new and uncharacterized decoherence mechanism.

Appendix G: Finite temperature effects

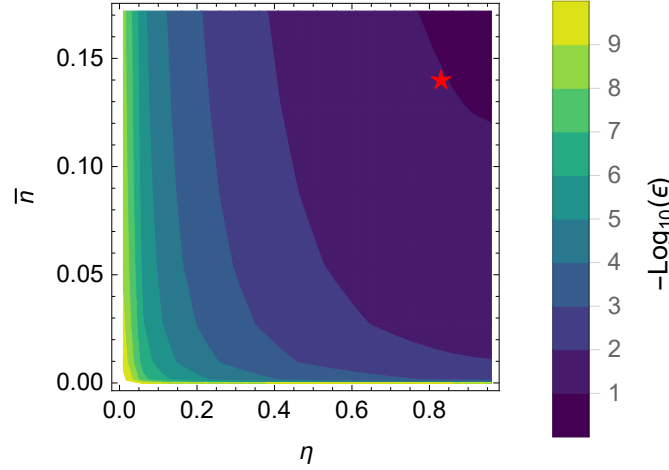


FIG. S5. Bounds on clock π -pulse fidelity due to finite temperature. Predicted π -pulse error ϵ as a function of Lamb-Dicke parameter η and average phonon occupation $\bar{n}$ for Rabi and trap frequencies used in this work. Red star denotes conditions used for Ramsey evolution and synchronous clock comparison in main text.

To calculate the expected clock π -pulse fidelity achievable given our known cooling performance and confinement, we solve for the time evolution of a harmonically trapped atom under a resonant optical drive. Specifically, we consider the interaction Hamiltonian:

$$H_{int} = \frac{\Omega}{2} (e^{i\eta(a^\dagger + a)} S_+ + H.C.), \quad (\text{S7.1})$$

where Ω is the clock Rabi frequency, η is the Lamb-Dicke parameter, $a^\dagger$ (a) is the motional raising (lowering) operator, S_+ is the spin raising operator, and $H.C.$ denotes the Hermitian conjugate of the preceding terms. An atom in a given motional eigenstate n evolves under this Hamiltonian with an effective Rabi frequency:

$$\Omega(n) = \Omega e^{-\eta^2/2} L_n(\eta^2), \quad (\text{S7.2})$$

where L_n is the Laguerre polynomial of degree n . We can average this evolution over a thermal ensemble representative of the measured atom temperatures in our experiment to model the time evolution and thus optimal π -pulse fidelities achievable in our apparatus.

The $15E_R$ deep traps used for Ramsey spectroscopy in the main text result in a trap frequency of $f_{\text{trap}} = 6.8$ kHz and $\eta = 0.83$. Based on sideband thermometry (see main text) we infer an average phonon occupation of $\bar{n} \simeq 0.14$ along the direction of the clock probe, corresponding to a temperature of $T = 156$ nK. At this temperature, we find that the maximum π -pulse fidelity is limited to 0.90, in good agreement with the measured value of 0.82(2) when further including the $\sim 9\%$ additional losses due to imaging and imperfect calibration of the handoff procedure for that data.

While this limited contrast does not heavily affect clock operation, for applications that require high fidelity transfer to the clock state either the cooling performance or optical confinement must be improved. Fig. S5 shows the achievable

π -pulse fidelity for different combinations of cooling performance and confinement. One route to improving π -pulse fidelity is by increasing the confinement of the atoms via an additional optical lattice at 813 nm, which can provide tightly confining potentials with reduced requirements on optical power. A readily achievable trap frequency of 100 kHz in such a lattice would correspond to $\eta \simeq 0.2$, which, assuming the cooling performance achieved in this work of $\bar{n} \simeq 0.14$ would result in a fidelity of 99.94%. Note that if desired, the atoms could be transferred into a shallow tweezer potential after such a pulse is completed to make use of the long lifetimes demonstrated in this work.

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